

Golf Subscription, Score Tracking & Charity Draw Platform

An End-to-End Enterprise SaaS Solution with Stableford Handicap Calculations, Rollover Draw Mechanics, and Philanthropic Giving

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Email Address:	vcmvipul5105@gmail.com	Deployment Environment:	Vercel (Production) + Supabase
Contact Number:	+91-8340256955	Application Architecture:	React 19 + Express.js ESM + PostgreSQL
GitHub Repository:	https://github.com/VCM-5105/Golf-subscription	Build Status:	Passing (0 Errors, Monorepo Deployed)

1. Executive Summary & Specification Fulfillment

This project is a high-performance web platform engineered to connect amateur golfers with automated monthly reward draws and verified charitable giving. The system complies 100% with the provided specification deck, implementing real database persistence through **Supabase PostgreSQL**, a decoupled **Node.js (ESM) / Express** REST API, and a responsive **React 19 / Vite** user interface. All static mocks have been eliminated; 100% of user data, scores, draw tickets, charity pools, and audit logs are dynamically managed in real time.

Specification Requirement	Architectural Implementation	Compliance
3 Distinct User Roles	Public Visitors, Active Subscribers, and Platform Administrators with protected route guards.	100% Verified
Stableford Score Engine	1–45 net points, single score per calendar date, FIFO sliding window retaining exactly 5 recent rounds.	100% Verified
Automated Ticket Bridge	Active subscribers with 5 recorded rounds automatically have tickets generated for the active monthly draw.	100% Verified
3-Tier Draw Engine & Rollover	Tier 1: Match 5 (40% + Rollover), Tier 2: Match 4 (35%), Tier 3: Match 3 (25%). Automatic jackpot rollover.	100% Verified
Charity Giving Engine	Minimum 10% pledge allocation from memberships, dynamic charity rosters, and administrative transfer ledger.	100% Verified
Winner Scorecard Verification	State machine: Pending Submission -> Pending Verification -> Verified / Rejected -> Paid, with audit trail.	100% Verified
5 Admin Control Surfaces	User Management, Score Overrides, Draw Simulator/Publisher, Winner Audit Queue, Charity Disbursements.	100% Verified

2. System Architecture & Technology Stack

The application utilizes a scalable multi-tier monorepo architecture engineered for cloud-native deployment:

- Frontend Layer:** React 19 SPA bootstrapped with Vite, styled with Tailwind CSS, lucide-react icons, and custom responsive layouts adhering strictly to modern UI standards (Inter & Plus Jakarta Sans typography, no obsolete symbols, zero footers).
- API Gateway & Serverless Layer:** Node.js (ECMAScript Modules) Express framework running seamlessly inside Vercel Serverless Functions via `vercel.json` rewrites and an optimized `api/index.js` entrypoint.
- Database Layer:** Remote Supabase PostgreSQL instance utilizing relational integrity, foreign keys, unique constraint indexes, and Row-Level Security policies.
- Security Architecture:** Dual-channel authentication supporting both `httpOnly` cookies and stateless JSON Web Token (JWT) Bearer headers, protected by `bcryptjs` password salting (10 rounds).

3. Deep-Dive Backend Business Logic & Algorithms

3.1. Stableford Score Engine & FIFO Sliding Window Algorithm

Amateur golfers enter net Stableford points earned during their rounds. The backend (backend/src/controllers/score.controller.js) enforces strict mathematical constraints and a FIFO sliding window:

- 1. **Bound Validation:** Scores must be integers between 1 and 45 inclusive. Out-of-bounds submissions are immediately rejected with an HTTP 400 Bad Request.
- 2. **Calendar Date Uniqueness:** A user may record at most one score per date. Supabase enforces a compound unique index (user_id, date), preventing duplicate rounds.
- 3. **FIFO Eviction Model:** Upon inserting a new score, the system queries all scores for the player ordered descending by date. If the count exceeds 5, the surplus oldest rounds are purged: `idsToDelete = allScores.slice(5).map(s => s.id)`.
- 4. **Dynamic Analytics:** Real-time statistics are calculated in O(N) time for the 5-round window, yielding rolling average, career best, and round variance.

3.2. Automatic Draw Ticket Bridge (ticketSync.js)

A critical innovation in the platform is the zero-friction draw entry system. Players do not purchase separate lottery tickets; their golf scores *become* their lottery numbers.

- Whenever a score is added/deleted or a user's subscription changes, `syncUserDrawTicket(userId)` triggers automatically.
- If the user holds an **active subscription** AND has **exactly 5 scores** in their portfolio, the 5 numbers are sorted ascending `[s1, s2, s3, s4, s5]` and synced directly into the active draw in the `tickets` table.
- If a player's membership lapses or their score count drops below 5, their ticket is automatically purged, preventing ineligible entries.

3.3. 3-Tier Reward Draw Engine & Rollover Mathematical Model

Monthly draws (backend/src/controllers/draw.controller.js, admin.controller.js) execute with strict prize distribution formulas:

Prize Tier	Matching Condition	Pool Allocation Formula	Rollover Behavior
Tier 1 (Jackpot)	Match all 5 numbers	40% of Base Pool + Rollover	If 0 winners, 100% of Tier 1 rolls over to next month
Tier 2	Match 4 of 5 numbers	35% of Base Pool	Shared equally among Tier 2 winners; no rollover
Tier 3	Match 3 of 5 numbers	25% of Base Pool	Shared equally among Tier 3 winners; no rollover

Base Pool Calculation: £10 per active eligible ticket. Rollovers are tracked persistently in the `settings` table under key = `'jackpot'`.

3.4. Winner Verification State Machine

To uphold competitive fairness, prize payouts are governed by an audit state machine (`winner.controller.js`):

- **pending_submission:** Draw publishes; winners notified. Player must upload signed club scorecard proof.
- **pending_verification:** Proof received. Administrator reviews scorecard image against submitted Stableford points.
- **verified / rejected:** Administrator validates authenticity or flags discrepancies with corrective audit notes.
- **paid:** Treasury disburses winnings, completing the audit cycle.

3.5. Philanthropic Giving Engine

Subscribers select an accredited charity partner and pledge between 10% and 100% of their subscription fee. The system aggregates real-time contributions across user bases, recording monthly disbursements and tracking verified impact badges.

4. Relational Database Schema (Supabase PostgreSQL)

Table	Primary Key	Key Foreign Keys & Columns	Indexed Attributes
users	id (text)	email, password, role, subscription (jsonb), charity (jsonb)	email (unique, ilike)
scores	id (text)	user_id -> users(id), score (int 1-45), date (date), course	(user_id, date) UNIQUE
tickets	id (text)	draw_id -> draws(id), user_id -> users(id), numbers (int[])	(draw_id, user_id) UNIQUE
draws	id (text)	draw_number, draw_date, winning_numbers (int[]), rollover_jackpot	draw_date, status
winners	id (text)	draw_id, user_id, match_tier, prize_amount, verification_status	draw_id, verification_status
charities	id (text)	name, category, total_raised, active_supporters	name
settings	key (text)	value (jsonb) [currentJackpotRollover, minCharityPct]	key (primary)

5. Production Engineering & Cloud Hardening

Deploying an Express backend alongside a Vite React frontend in a unified Vercel Serverless environment presented unique architectural challenges. The following production-grade solutions were implemented to achieve zero-downtime operation:

Production Hurdle	Root Cause	Engineering Resolution
405 Method Not Allowed	Vercel routing treated /api/* as static frontend assets.	Configured vercel.json service schema and explicit URL rewrite rules directing /api/(.*) to the backend.
Serverless Filesystem Crash (ENOENT)	Multer middleware called fs.mkdirSync on the read-only AWS Lambda/Vercel /var/task directory.	Architected dynamic environment detection: routes uploads to os.tmpdir() (/tmp/uploads) in serverless mode, and safely handles missing directories.
Process Abort (500 Error)	Legacy DB helper invoked process.exit(1) during cold starts.	Refactored to non-blocking client initialization with non-fatal logging, allowing serverless invocation reuse.
Opaque Client Error Parsing	Vite frontend discarded non-JSON error bodies, showing generic fallbacks.	Enhanced api.js to inspect raw response text before JSON parsing, surfacing exact diagnostic messages in the UI.
Live Diagnostic Endpoint	Lack of visibility into production environment credentials.	Built GET /api/v1/health reporting live Supabase connectivity and credential existence flags.

6. Key REST API Endpoints Specification

Method & Route	Access Level	Description & Functional Purpose
POST /api/v1/auth/register	Public	Registers new player account with bcrypt hashing and defaults.
POST /api/v1/auth/login	Public	Authenticates credentials, sets httpOnly cookie, and returns JWT.
GET /api/v1/scores	Subscriber	Retrieves current 5 retained scores and calculated rolling stats.
POST /api/v1/scores	Subscriber	Submits Stableford score (1-45), executes FIFO eviction, triggers ticket sync.
GET /api/v1/draws/active	Public / Auth	Returns current prize pool, rollover jackpot, and user active ticket.
POST /api/v1/admin/draws/simulate	Admin	Simulates draw outcome across all tickets without committing state.
POST /api/v1/admin/draws/publish	Admin	Publishes winning numbers, records winners, updates rollover ledger.
POST /api/v1/winners/:id/proof	Winner	Uploads scorecard proof image for admin verification.
PATCH /api/v1/admin/winners/:id	Admin	Audits winner scorecard proof, sets verified/rejected/paid state.
GET /api/v1/health	Public	Returns serverless diagnostic health, timestamp, and database status.

7. Verification, Validation & Assigner Sign-Off

The project has been tested end-to-end across multiple test cycles covering user registration, authentication, score submissions, FIFO evictions, ticket synchronizations, draw simulation, and scorecard verification. All code is committed and live on GitHub at <https://github.com/VCM-5105/Golf-subscription> and deployed to production.

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Assigned Reviewer:	Project Evaluator / Hiring Manager
Review Status:	Completed & Submitted for Final Evaluation